

Smart City IoT Application for Road Infrastructure Safety and Monitoring by Using Digital Twin

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Abstract—Industrial internet of things (IIoT) enhances automation and safety in transportation flow and new technology must meet the agents. Digital twins in road infrastructure will improve public safety and security remotely in real time and train the infrastructure for future self-automation systems. This study incorporates the notion of digital twin, adopts the learning and perception model of virtual space, and investigates essential technologies for linking to system infrastructure's physical area. This article examines high automation and intelligent control in the data center, from planning to maintenance to virtual space optimization. Physical space sensors capture data and share it with virtual space via 5G, Wi-Fi, etc. Cloud servers enable end users visualize data and information operations with digital models and mobile apps.

Index Terms—Digital twin; IIoT; WIFI; 5G; ISO/APK

I. INTRODUCTION

The unpredicted increasing travel demands mean urban areas' transportation systems operate about every hour a day. During the seasonal and weekend demands on the nation's traffic infrastructure, which increases transportation capacity [1], [2]. Based on these mentioned drawbacks in traditional traffic monitoring systems, there is a need to improve the traffic management system by using intelligence technologies such as sensors, which reduce traffic jams, road accidents, and injuries as well as reduce travel time. The old monitoring system was made a few decades ago based on the number of vehicles and small traffic agents, but as IOT got better, it found that the number of vehicles and roads had grown by a lot [3]–[5]. Due to the need for smarter solutions, which have been used recently and require new technologies to adopt intelligent transport systems, intelligent traffic system (ITS) is

able to enable logistic transportation systems, and the adoption of autonomous vehicles (AVs) has brought huge improvements to traffic control systems [6]–[8]. However, the unprecedented increase in data has created significant challenges for ITS developers. The deployment of vehicle-based sensors for a variety of applications has resulted in an increase in data production and collection in ITS [9]. Analyzing and understanding sensor data [10], [11] and figuring out which public transportation systems have few delays is a big problem that needs quick and reliable solutions [12], [13] to meet efficiency and quality of service (QOS). Vehicle-to-vehicle communication (V2V) is technology-based research that allows information to be shared from one vehicle to another. Radar [14]–[16] assists the vehicle in recognizing the agents and their location in the environment. The development of ITS through the use of IOT applications such as the internet of vehicles (IOV) allows vehicles-to-everything communication (V2X) [17], [18]. Smart traffic lights and other sensors, such as global positioning system (GPS), lidar, image processing, audio detection, and cameras for visual objects [19], [20], aid in the collection of ITS perception data. The whole mechanism of transforming the systems from TTS to ITS is shown in Fig. 1.

Data collection by human beings and manual sharing [21] and sending this data by mail box or fax takes a long time, and after the upgrade of this traditional traffic system (TTS) to ITS, collecting transmission and control is fully automated, which is a more efficient way to manage traffic in real time. ITS provides rapid and real-time data transmission between the road and control room [22]. Intelligent transport systems (ITS) help to manage and control traffic more effectively. The major reasons for increasing traffic control issues are: 1) A rise in the number of cars on the road; 2) Time for road intersections; 3) Miss is in charge of distribution traffic and

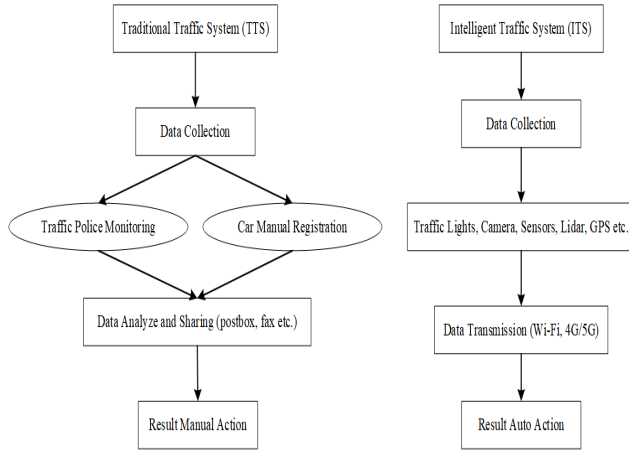


Fig. 1. Scheduling of Tasks and VMs in cloud computing

vehicle routes; 4) There is no communication between V2I and vehicle-to-pedestrian (V2P) vehicles, which also makes traffic jams worse; 5) Traffic system that is not fully or partially automated; 6) Failure to give priority to emergency vehicles (ambulances, firemen, police, and so on). DT-based models in traffic systems contribute to improving the intelligence level, efficiency, and safety and security systems [23], [24]. DT-based systems are also able to be automatically reconfigured and auto-upgraded by themselves. In recent years, the Digital Twin (DT) technology has become more useful for combining physical and virtual models with a set of model data [25]–[27] to simulate physical systems with the following features:

- DT can be used to simulate and describe a physical object’s properties over its life cycle.
- DT includes virtual and physical space, and can realize data interaction and real-time mapping between them by using middle ways.
- DT is used not only to describe the properties and behavior of physical space, but also to provide physical system solutions.

II. RELATED WORKS

The focus of this article is to cover the number of sensors in a vehicle used to collect data (as shown in Table.1) and the DT used for data transfer. Short, medium, and long-range sensors are used depending on the transmission range of the wireless technology. Intelligent Transport System (ITS) applications require a large number of sensors to assist in automation and communication. [28] demonstrates the most recent developments and advancements in smart vehicle perception and sensor technology. It describes the Dedicated Short-Range Communications (DSRC) and cellular vehicle-to-everything (C-V2X) resource allocation schemes. But both technologies can only be used for medium- and long-range communications, and they have problems like applications that take too long [29]. This paper contributes to reducing latency with long-range communication by implementing a 5G network. For short-range communication with a transmission distance of about 20 m, multiple technologies such as Bluetooth [30],

infrared, ZigBee [31], and ultra-wideband (UWB) [32] have been proposed. Several researchers proposed features and key characteristics such as wireless technology, power consumption, and data rate and transfer time. Long-term evolution-advanced (LTE-A) pro with cellular-V2X capability, as well as 3GPP key technologies for 5G-NR, have been released [33]. These can be used for long-distance communication of 100m or more. These technologies have a large coverage area, high data rates, and a low latency rate of 50–80 ms, depending on how busy the cellular network is. However, they are not good for safety-critical applications [34].

III. FRAMEWORK

III. FRAMEWORK A “digital twin”, a virtual space of a physical asset, is created as a digital replica of a machine, equipment, or physical site. These digital assets can even be created before the asset is physically built. To create a digital twin of a physical asset, engineers collect and integrate data from a variety of sources, including physical data, sharing data, operational data, and analytical data. All of this information and AI algorithms are integrated into physical-based virtual models that you can apply analytics to get the latest information about your physical assets. A consistent flow of data will help you get the best analytics and asset information to help you optimize your results, as shown in Fig. 2. Gateways such as 5G, WIFI, etc. enable physical space to share collected data with virtual space. Middleware like Amazon Web Services (AWS), Google, Huawei, and Baidu are cloud-based data centers that offer virtual management, data storage, and function execution services over the network. Android apps can be used to easily operate and control DT-based ITS, share information about vehicles, people, objects, and infrastructure, and collect taxes and fines online in real time. Our gateway layer is made up of cellular 5G technology, which can provide a very high latency rate with very little delay shown in Fig. 1.

IV. METHODOLOGY

The main model of ITS consists of communication, data collection, and operation as shown in Fig.3. RSU is in charge of gathering all information from real-world environments via cars, infrastructure, and local boards, with cars also having local boards for communication.

A. Manual Data Collection

It is used as a traditional technique to collect data about traffic, such as vehicle numbers, models, pedestrians, and traffic light manual operations. This process is done by human resources to manage this data.

B. Specifications Data by Infrastructure based technology

Local sensors (shown in Table. 2) are used to collect data and information about agents (vehicles, pedestrians, road infrastructure, obstacles, etc.). Which are

- Traffic light detectors: Intelligent traffic light detectors are

TABLE I
RELATED WORK

Ref.	Technology	Achievements	Range
[35] [36]	Network protocols for transport and physical layer, IoT	the security aspects for V2V communication	Long range
[37] [38]	ZigBee, Infrared	Vehicle Identification and validation, improve security	Short range
[39]	Ultra-wideband (UWB)	High bandwidth Low energy implements for validation and collision alert system to save life	Medium and long range
[40]	DSRC and LTE	Cellular vehicle networks	Medium and Long range
[41]	3GPP	D2D, C-V2V	Long and Short Both
[42]	Bluetooth	RSU Detection and Validation high speed vehicle	Long and Short Both
[43]	Sensors such as Camera, Lidar, GPS, Ultrasonic etc..	Data Perception and planning	Short and long range
[44]	IEEE 802.11bd and 5G-NR, DSRC, C-V2X	Validation, safety and security, the suitability of communication	Long range
[45] [46]	LTE-V2X, NR-V2X, C-V2V	Road safety and infrastructure, testing and validating, low-latency, high-reliability, and high-throughput communications for various C-V2X applications	Long range

made up of a number of local sensors that are used to collect data and keep track of how many cars and people are coming and going.

- Image detectors: Intelligent cameras are used to scan vehicle numbers, times, speeds, and locations, as well as to get all kinds of information about the driver, like pictures, videos, audio recordings, gestures, etc.
- RSU (Road Side Unit): Basically, an RSU is used to monitor the road infrastructure by detecting cameras, traffic lights, and other sensors. The RSU is responsible for storing and transmitting detected data directly to the vehicle and control room.

C. Data by vehicle Based technology

Data collection from vehicles: Local sensors make vehicles able to collect more efficient data to improve safety and transport flow. As shown in Fig. 4, vehicle data can be divided into external road environment data collected by sensors, such as location, weather, and roadside conditions or repairs, other behavioral data about agents on the road, and maintenance information. Data from the driver's Internet of Things (IoT) devices, such as phones, vehicle-based Android devices, etc., is also part of the vehicle's internal data. These devices include phones, vehicle-based Android devices, etc.

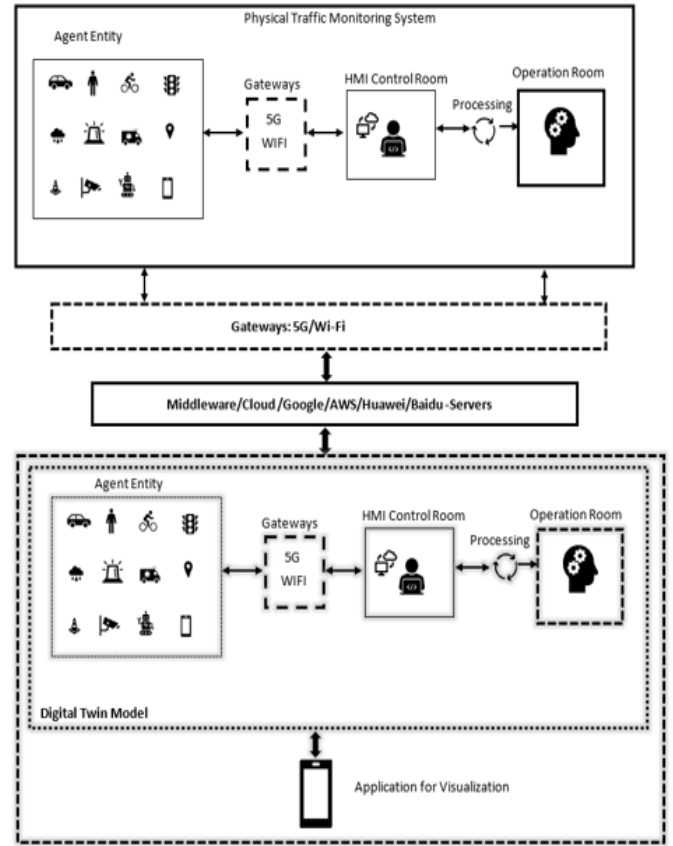


Fig. 2. Physical spaces to virtual space

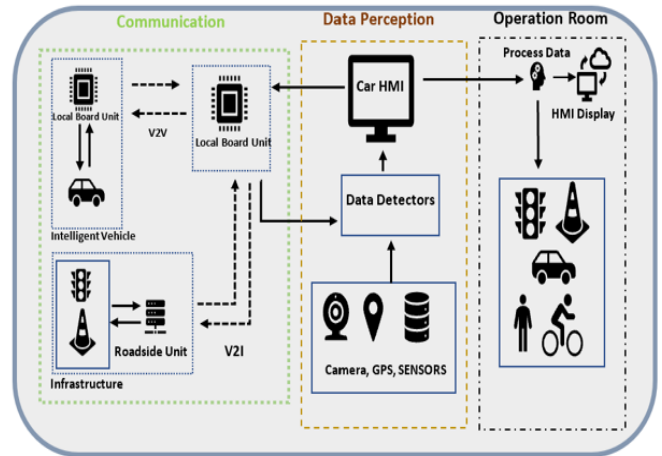


Fig. 3. Intelligent monitoring system

V. RESULTS AND DISCUSSION

A. Results

In the Intelligent Transport System, data collection, use, and processing are becoming more and more important. Modern vehicles are always busy collecting information from their surroundings. Today's vehicles are linked via 4G/5G, WIFI, GPS, etc. to share information with other vehicles (V2V),

TABLE II
SENSOR AND THEIR FEATURES

Sensor	Features
Camera	To monitor and detect obstacle, behavior of agents on road
Lidar	Work as road line scanner, identification agents and distance range detection, etc.
Health Monitor	Device is to monitor driver health, heartbeat, temperature and alcoholic level [47]
GPS	Location, Navigation, Timing, Monitoring, Mapping
Optical Communication Image	Optical signal is used to connect Vehicle for faster data rate in V2V up to 54 Mb/s [48]
Fuel level sensor	To indicate fuel levels
Ultrasonic Sensor	Object detection, high accuracy detection, distance range measure
Radar	To Estimate velocity, distance and angle of detected objects [49]
Image Sensor Communication	visualization and analyze image data, visible light communication

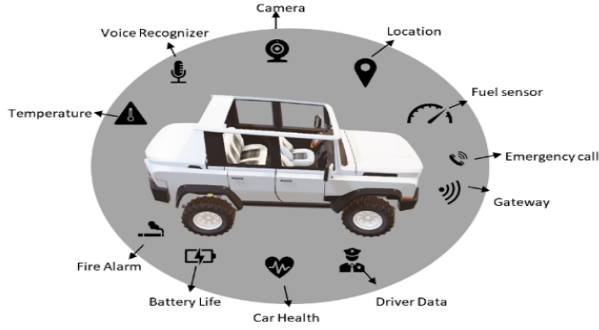


Fig. 4. Vehicle based sensors for collecting data

the cloud (V2C), pedestrians (V2P), infrastructure (V2I), and everything (V2X) as shown in Fig. 5. Fig. 6 shows a Unity 3D simulation with vehicles that automatically share information with each other and work together to find people, barriers, networks, and infrastructure.



Fig. 5. Vehicle to everything data sharing

B. Discussion

Cellular Network 5G/WIFI: The communication-based Cellular Network 5G collects data such as the number of users, offline or online users, their history, machine-to-machine connectivity, user identification, and health records. This data is then broadcast to other users. As shown in Fig. 7, we use our network to share all information for improving perception purposes.



Fig. 6. Unity simulation, where shows connected vehicles, V2V, V2I, V2P and V2N.

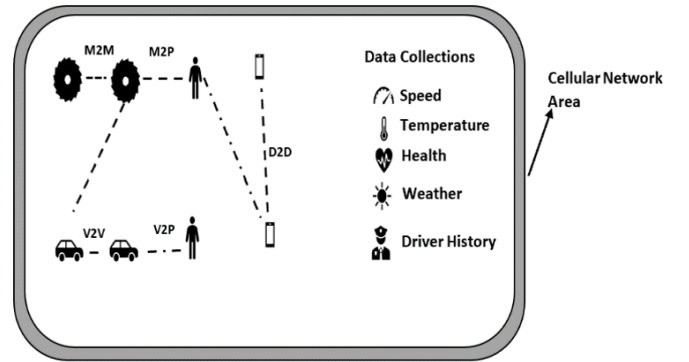


Fig. 7. Data collection and sharing by cellular network

GSM: uses GPRS (General Packet Radio Services) to collect data about positioning and real-time location and easily transmit data. As shown in Fig. 8, the latest GPRS has features to collect data about:

1. Realtime Tracking
2. Notification of an Event
3. Log of Historical Path Data
5. Overspeed Warning
6. Journey Distance Accumulation
8. Immobilization of a vehicle from a remote location

The continued development of DT shows that DT will continue to play an important role in the IIoT. Despite the growing demand for digital twins, we need to study how to integrate individual digital twins into our enterprise network in the future. A concept has been put forward in this paper, based on the ITS, where a DT system is made. The ITS (Intelligent

Transportation System) is used as the digital twin software platform to establish the traffic control and monitoring model, and an online data collection DT system has been developed to achieve real-time monitoring and mapping between physical and virtual space. Finally, the DT system developed is verified to be highly flexible, real-time and secure. DT is a pure virtual system that synchronizes and directs real-time connections to a physical system. Cloud servers such as Google, AWS, Baidu, and Huawei can be accessed to provide storage for virtual space. In the future, expect to see the expansion of the IoT, and with it, some version of digital twin technology. In the future, we should expect the expansion of the Internet of things and its accompanying versions of digital twin technology. By 2022, 85% of IoT platforms are expected to include some sort of digital twin. As more and more products in the workplace become "smart," or connected to the Internet so they can be accessed and controlled remotely, digital twin technologies will also become more widely available. These twins allow remote control of smart homes and smart city monitoring systems.

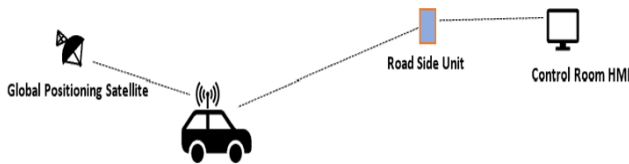


Fig. 8. GPRS data collection and transmission

VI. CONCLUSIONS

DT technology has been used to track and control transportation systems (MCTS) online and in real time. A multi-level layer model method for real-time monitoring, communication, and real-time synchronization between physical space and virtual space has been discussed in detail. The main contents of this paper can be summarized as follows: (1) Using sensors, cameras, and network technologies, model the digital twin's virtual space as identical to the physical ITS system. (2) The framework of the DT communication platform for the ITS process is made so that data can be collected and understood in real time between the virtual and real worlds. The real-time data sharing process of the proposed framework has been defined. (3) A DT system that employs 5G technology to enhance real-time instruction and target tracking while maintaining a high level of safety and security.

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